

Degenerate Forward Maps and Fisher Holes

Blind Sector Health, Exterior–Interior Degeneracy, Transport Holonomy, and Chaotic Observability

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Abstract

Many inverse problems in gravitational and field-theoretic settings are structurally degenerate: distinct interior sources, histories, or parameter values can induce almost indistinguishable exterior observations. This paper studies that issue at the level of finite forward maps and observational Fisher geometry. Given a forward map $y = f(\theta)$ with Jacobian J , the Fisher form $F = J^T \Sigma^{-1} J$ defines a resolved sector and a null or near-null “hole.” We define a Fisher health vector consisting of numerical rank, hole dimension, effective rank, resolved condition number, smallest relative eigenvalue, and eigenvalue clustering. Theorems state the elementary but important obstruction: directions in $\ker J$ cannot be recovered from the observation, and perfect residual fit does not imply parameter identifiability. Numerical batteries then show: (i) ordinary residual and cross-validation diagnostics can remain excellent while the Fisher hole expands; (ii) a radial exterior-observation toy has many strongly different interiors with essentially identical exterior data; (iii) chaotic dynamics with degenerate observation produces exact or near-null directions; and (iv) transport around near-degenerate Fisher fibers can have small local leakage but nontrivial global holonomy. The contribution is not a method for recovering unobserved information. It is a method for exposing which directions are resolved, which directions lie in the hole, and where fit-based confidence is unsafe.

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1 Purpose

Let

$$f : \Theta \rightarrow Y, \quad y = f(\theta),$$

be a finite forward map. In a gravitational interpretation, θ may represent matter, stress, or interior-profile parameters, while y represents exterior or finite-resolution observations. The central problem is not noise alone. The harder problem is structured degeneracy:

$$\theta_1 \neq \theta_2, \quad f(\theta_1) \approx f(\theta_2).$$

The purpose of this paper is to measure this degeneracy geometrically using the Fisher form

$$F = J^T \Sigma^{-1} J, \quad J = \frac{\partial f}{\partial \theta}.$$

Non-claim 1.1 (No recovery from the hole). If a direction is unobserved by the forward map, this paper does not claim to reconstruct it. The claim is only that the unresolved sector can be identified and separated from the resolved sector.

Non-claim 1.2 (No full GR inverse theorem). The exterior/interior example below is a radial moment toy. It is designed to mimic the structural fact that different interiors may induce similar exterior data. It is not a theorem about the full Einstein constraint equations.

Non-claim 1.3 (No claim that residual diagnostics are useless). Residual, validation, and R^2 diagnostics measure fit quality. The claim is narrower: they do not, by themselves, measure identifiability of the parameter directions.

2 Fisher holes and health vector

Definition 2.1 (Fisher spectrum and resolved rank). Let $F \succeq 0$ have eigenvalues

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0.$$

For a relative threshold $\tau > 0$, define

$$r_\tau = \#\{i : \lambda_i / \lambda_1 > \tau\}, \quad h_\tau = n - r_\tau.$$

The resolved sector is

$$\mathcal{R}_\tau = \text{span}\{v_i : \lambda_i / \lambda_1 > \tau\},$$

and the hole is

$$\mathcal{N}_\tau = \mathcal{R}_\tau^\perp.$$

Definition 2.2 (Health vector). The Fisher health vector is

$$\boxed{\mathcal{H}(F; \tau) = (r_\tau, h_\tau, r_{\text{eff}}, \kappa_\tau, \lambda_n / \lambda_1, c_{\text{cluster}}),}$$

where

$$r_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}, \quad \kappa_\tau = \frac{\lambda_1}{\lambda_{r_\tau}},$$

and c_{cluster} counts adjacent resolved eigenvalues whose relative gap is below a declared cluster threshold.

3 Elementary obstruction theorems

Theorem 3.1 (Null directions are observationally indistinguishable). *Let $f(\theta) = J\theta$ be linear. If $v \in \ker J$, then*

$$f(\theta + av) = f(\theta)$$

for every scalar a . Therefore no estimator using only $y = f(\theta)$ can distinguish θ from $\theta + av$.

Proof. Since $Jv = 0$,

$$f(\theta + av) = J\theta + aJv = J\theta = f(\theta).$$

The observation is identical for all a , so the distinction is not present in the data. $\square$

Corollary 3.2 (Perfect fit does not imply identifiability). *If $y \in \text{Im } J$, then there exists $\hat{\theta}$ such that $J\hat{\theta} = y$. Hence the residual can vanish even when $\ker J \neq \{0\}$.*

Theorem 3.3 (Effective rank detects spectral collapse). *If F has one dominant eigenvalue and all remaining eigenvalues approach zero, then*

$$r_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2} \rightarrow 1.$$

Proof. Let $\lambda_1 > 0$ and let $\sum_{i>1} \lambda_i/\lambda_1 \rightarrow 0$. Then both numerator and denominator are asymptotic to λ_1^2 , so the quotient tends to 1. $\square$

4 Battery I: standard diagnostics versus Fisher health

A structured linear forward map was built with controlled singular-value collapse. Noisy observations were fitted by pseudoinverse and ridge estimators. The table compares ordinary fit diagnostics with Fisher health.

ϵ	R_{LS}^2	residual	CV rel. MSE	Fisher rank	hole dim.	eff. rank	null output rel.
10^{-1}	0.999999	8.63×10^{-4}	1.45×10^{-6}	6	14	2.304	1.26×10^{-14}
10^{-2}	0.999999	8.98×10^{-4}	1.08×10^{-6}	6	14	1.377	3.69×10^{-14}
10^{-3}	0.999998	1.51×10^{-3}	7.84×10^{-6}	5	15	1.135	5.10×10^{-13}
10^{-5}	1.000000	5.29×10^{-4}	3.73×10^{-7}	3	17	1.020	5.74×10^{-11}
10^{-8}	0.999347	2.55×10^{-2}	1.31×10^{-3}	2	18	1.001	3.41×10^{-6}
exact drop	0.999998	1.48×10^{-3}	2.67×10^{-6}	2	18	1.000	8.23×10^{-11}

Finding 4.1 (Residual health separation). Residual, R^2 , and cross-validation diagnostics remain excellent in regimes where the Fisher hole has dimension 17–18 and the effective rank is approximately one. These diagnostics certify fit quality, not identifiability.

5 Battery II: exterior/interior degeneracy toy

A radial interior profile

$$\rho(r) = \sum_{i=1}^{24} \theta_i b_i(r)$$

was observed through five exterior moment-like probes. The Jacobian therefore maps a 24-dimensional interior basis to five exterior observables.

basis dim.	obs. dim.	rank	hole dim.	eff. rank	cond.	profile change	exterior change	R^2
24	5	5	19	1.072	1.59×10^7	2.098	2.39×10^{-13}	1.0

Finding 5.1 (Exterior shadow degeneracy). In the radial toy, a perturbation can change the interior profile by more than twice its original norm while changing the exterior observation by only 2.39×10^{-13} relative error. The fit is perfect, but the Fisher hole has dimension 19.

6 Battery III: chaos with exact and near-rank degeneracy

Lorenz dynamics was used as a chaotic generator. The unknown parameter is the initial condition $x_0 \in \mathbb{R}^3$. Different observation maps were applied to the trajectory, producing exact rank drops and near-degenerate Fisher spectra.

Observation	obs. dim.	rank	hole dim.	eff. rank	cond.	null output	strong output
one sample x	1	1	2	1.000	1.0	3.97×10^{-5}	2.75×10^{-3}
two samples x	2	2	1	1.003	677.5	6.22×10^{-7}	1.56×10^{-2}
radius sparse	4	3	0	1.001	3.15×10^6	3.21×10^{-5}	6.36×10^{-2}
$x + y$ sparse	8	3	0	1.003	3.81×10^4	1.67×10^{-4}	3.21×10^{-2}
full short	30	3	0	1.096	102.0	3.44×10^{-3}	3.47×10^{-2}

Finding 6.1 (Chaos does not remove observational holes). Chaotic dynamics can amplify resolved directions, but deliberately sparse or degenerate observation maps still create exact null directions or severe near-null directions. The Fisher health vector distinguishes full-rank but ill-conditioned observation from actual rank loss.

7 Battery IV: controlled transport and holonomy

A loop of Fisher forms was constructed with a two-dimensional resolved sector. The resolved sector was transported around loops with controlled coupling to near-null directions. The table separates no-coupling, weak-coupling, and strong-coupling cases.

mix	null gap	max step angle	mean step angle	final subspace angle	internal holonomy
0.00	0.50	7.44×10^{-16}	1.89×10^{-16}	0	4.21×10^{-8}
0.00	0.02	7.44×10^{-16}	1.89×10^{-16}	0	4.21×10^{-8}
0.15	0.50	5.85×10^{-3}	5.70×10^{-3}	0.300	0.035
0.15	0.02	5.85×10^{-3}	5.70×10^{-3}	0.300	1.571
0.30	0.50	1.17×10^{-2}	1.13×10^{-2}	0.600	1.571
0.30	0.02	1.17×10^{-2}	1.13×10^{-2}	0.600	1.571

Finding 7.1 (Local stability does not imply trivial global transport). With no coupling to the hole, both final subspace angle and internal holonomy are numerically zero. With coupling to near-null directions, local step angles remain small but global subspace displacement and internal holonomy become nontrivial. Therefore transport diagnostics should be separated into local leakage, final subspace angle, and internal frame holonomy.

8 Interpretation for blind sector geometry

The batteries support the following operational rule:

$$\text{fit diagnostics} \neq \text{identifiability diagnostics}.$$

The resolved sector is the part of parameter space supported by the Fisher spectrum. The hole is not failed optimization; it is the set of directions that the observation map does not distinguish at the declared threshold.

Corollary 8.1 (Dangerous-fit criterion). *A fit is structurally unsafe if residual diagnostics are good while h_τ is large, r_{eff} is small, or λ_n/λ_1 is near zero. In that case, the fit may be accurate in observation space while unreliable in parameter space.*

9 Conclusion

This paper establishes a conservative degeneracy result. No method can reconstruct directions absent from the observation map. The blind-sector health construction does not solve that impossibility. Its role is to prevent false confidence: it exposes the resolved Fisher sector, measures the hole, detects rank collapse and near-null directions, and separates local transport leakage from global holonomy. The exterior/interior toy and chaotic observation tests show that this distinction is not cosmetic. It is necessary whenever many hidden states can cast nearly the same observable shadow.

A Reproducibility

All numerical results reported in this paper were generated using deterministic random-number generators with explicitly fixed seeds. The primary outputs are stored in

- `rg_degeneracy_tightening_battery_results.json`,
- `rg_degeneracy_tightening_battery_summary.txt`,
- `rg_degeneracy_hole_stress_battery_results.json`.

The batteries use only standard finite-dimensional linear algebra, deterministic numerical optimization, and explicitly seeded stochastic components. The following seeds were used:

- Structured forward-map and Fisher-degeneracy batteries:

```
rng = np.random.default_rng(20260728).
```

- Cross-validation splits:

```
random_state = 7.
```

- Exterior/interior degeneracy toy: generated from the same master generator

```
np.random.default_rng(20260728).
```

- Blind-sector and reduced-cosmology batteries:

```
np.random.default_rng(20260727)
```

for observation probes and

```
np.random.default_rng(42)
```

for envelope-identity probe locations.

- Random-sector baselines:

$$\text{seed} = 100, 101, \dots, 109$$

for the reduced-cosmology comparison and

$$\text{seed} = 123$$

for the blind-sector observability baselines.

- Observation-map perturbation tests:

$$\text{seed} = 2,$$

with perturbation amplitudes reported in the corresponding battery.

- Fisher-hole transport and holonomy batteries: deterministic, with no stochastic component once parameters are specified.
- Lorenz chaos batteries: deterministic numerical integration with fixed initial condition

$$x_0 = (1, 1, 1),$$

and fixed finite-difference step

$$h = 10^{-5}.$$

Consequently, rerunning the supplied scripts with the same numerical libraries, thresholds, and seed values reproduces the reported spectra, ranks, hole dimensions, effective ranks, transport angles, and envelope-identity diagnostics up to ordinary floating-point roundoff.

References

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- [3] E. N. Lorenz, “Deterministic nonperiodic flow,” *Journal of the Atmospheric Sciences* 20 (1963), 130–141.